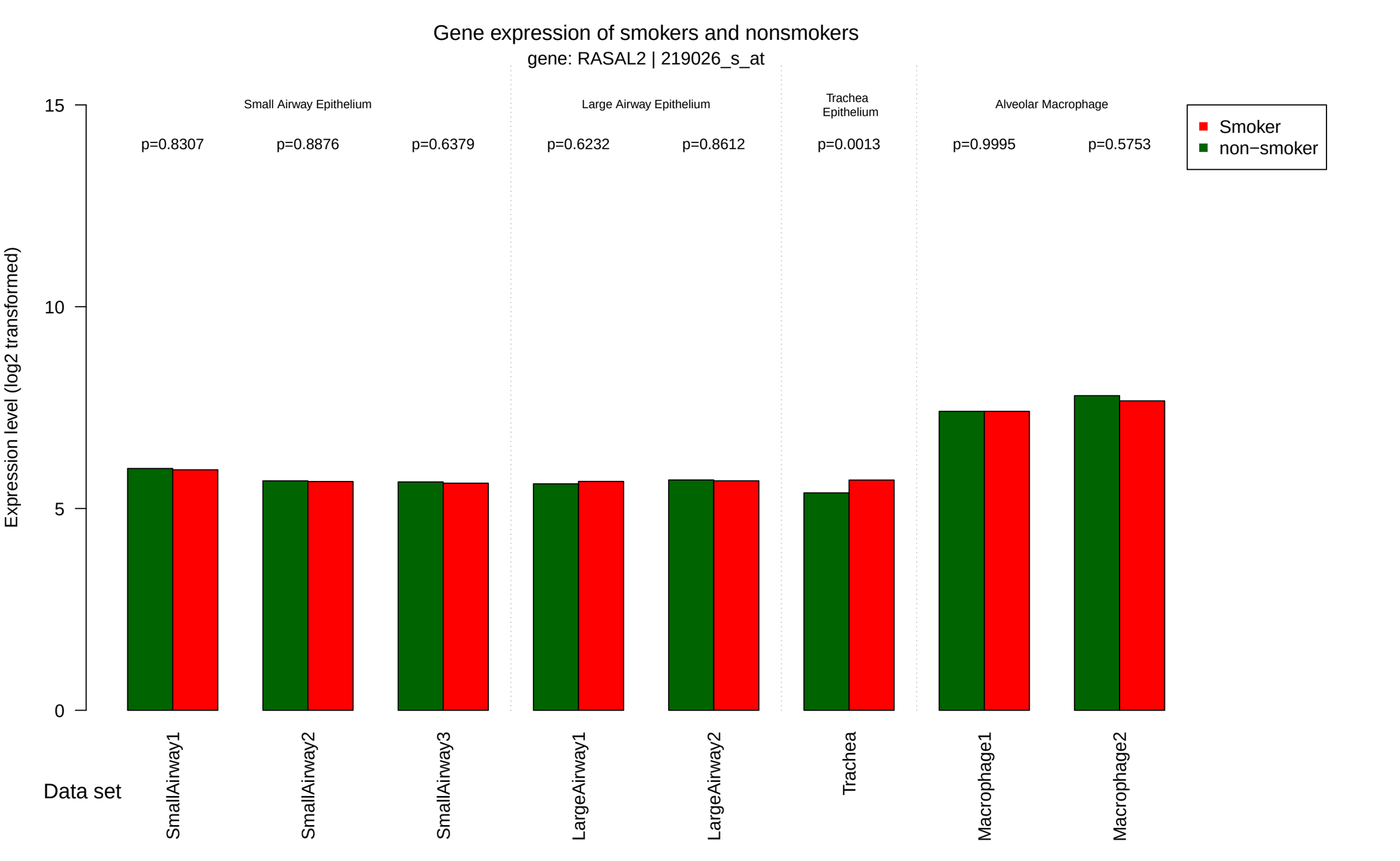
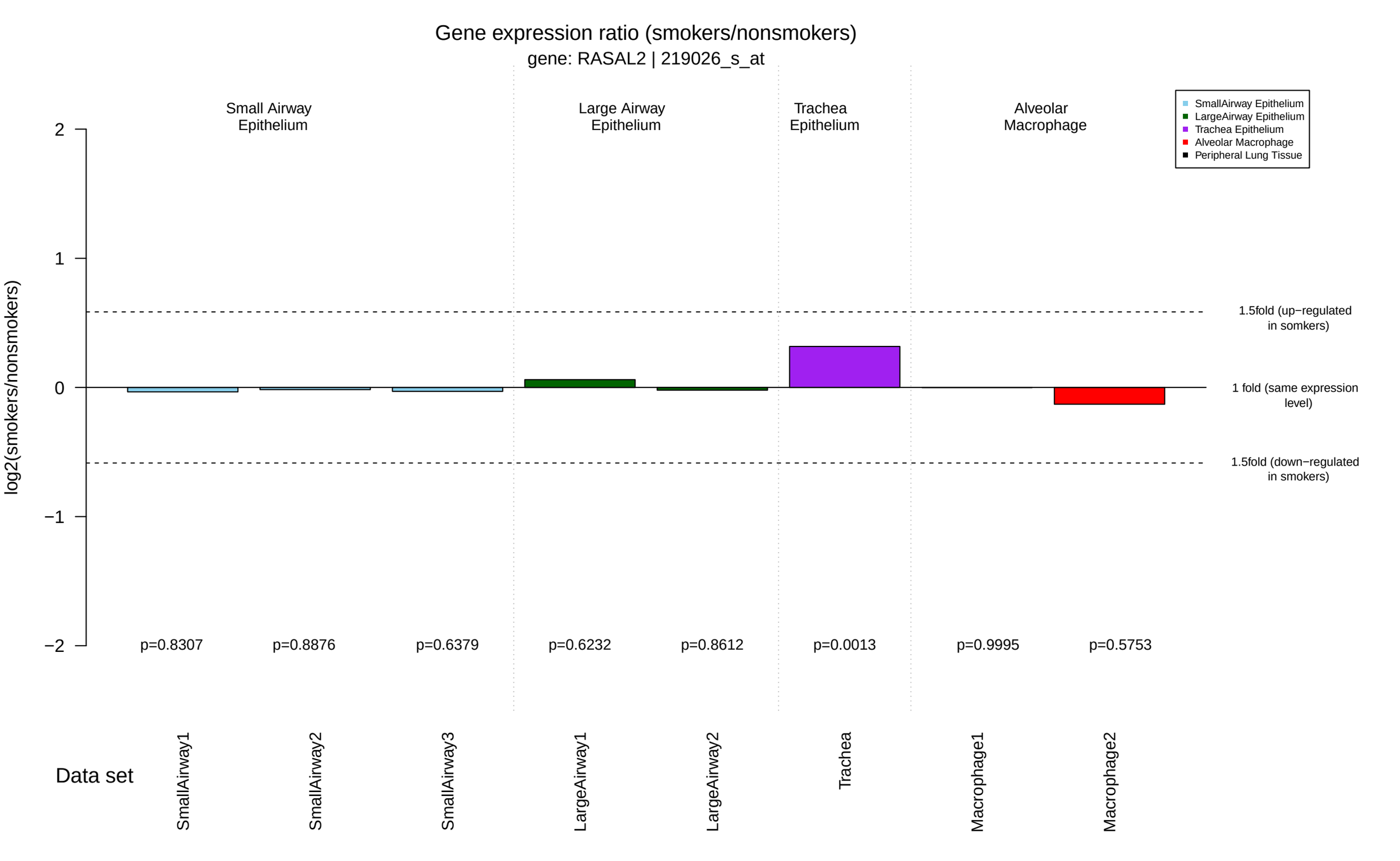
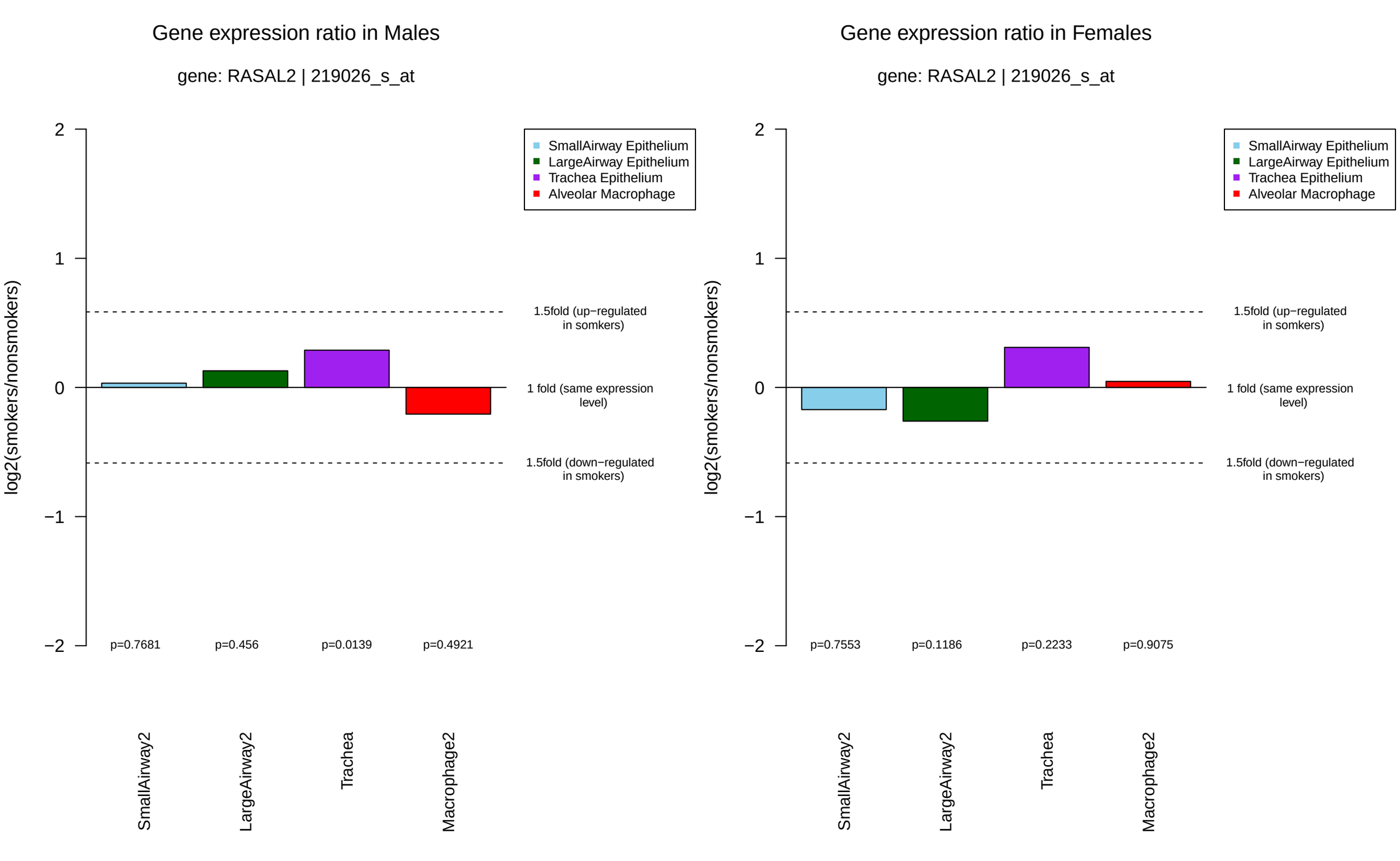
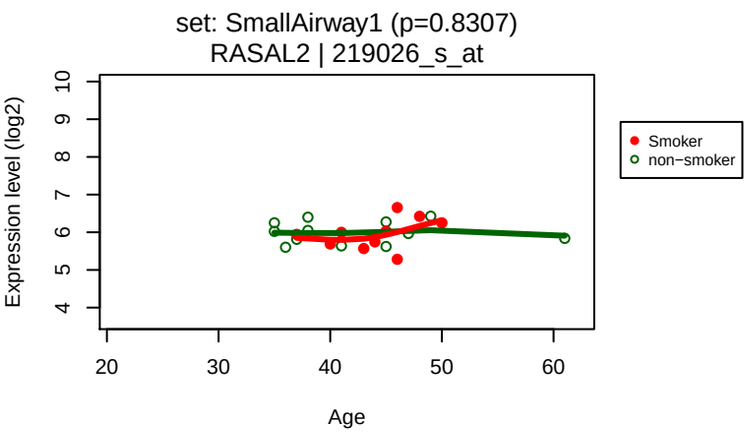
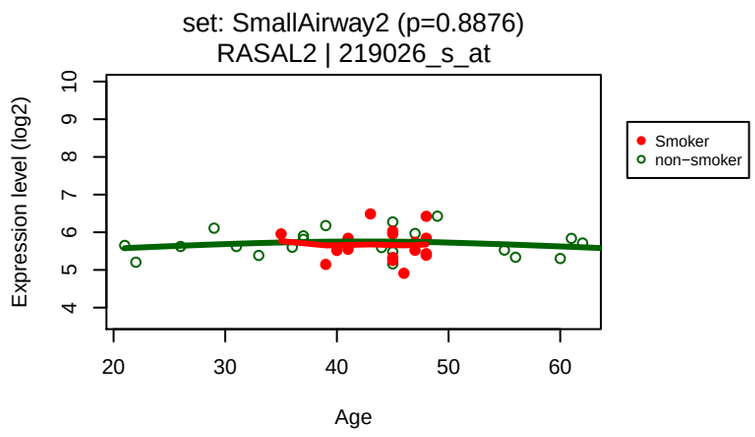
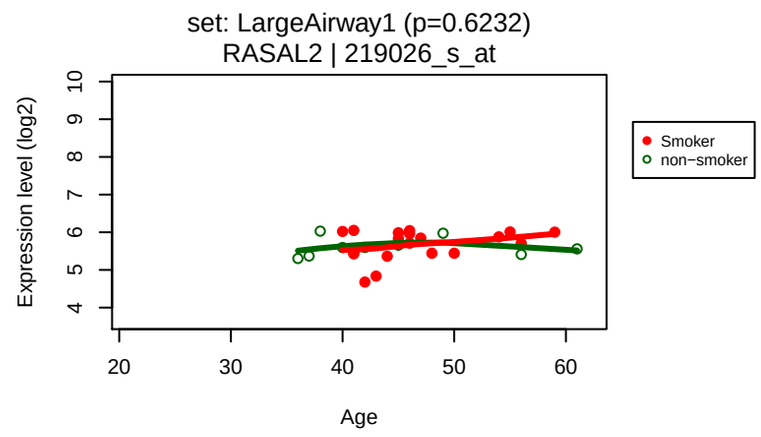

Gene expression of smokers and nonsmokers
gene: RASAL2 | 219026_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8307
p=0.8876
p=0.6379
p=0.6232
p=0.8612
p=0.0013
p=0.9995
p=0.5753
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

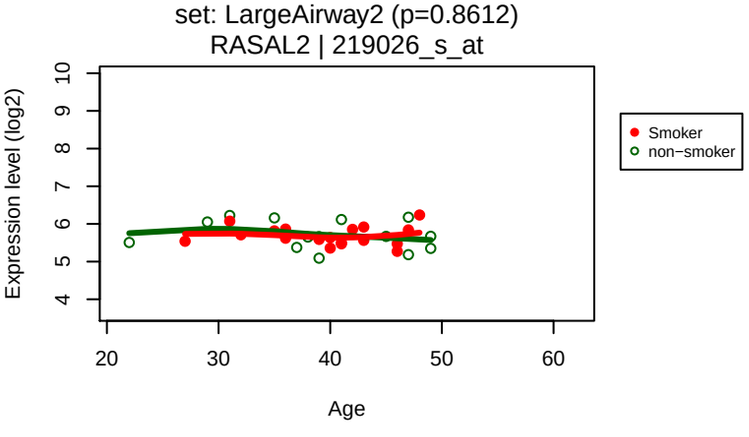

Gene expression ratio (smokers/nonsmokers)
gene: RASAL2 | 219026_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8307
p=0.8876
p=0.6379
p=0.6232
p=0.8612
p=0.0013
p=0.9995
p=0.5753
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

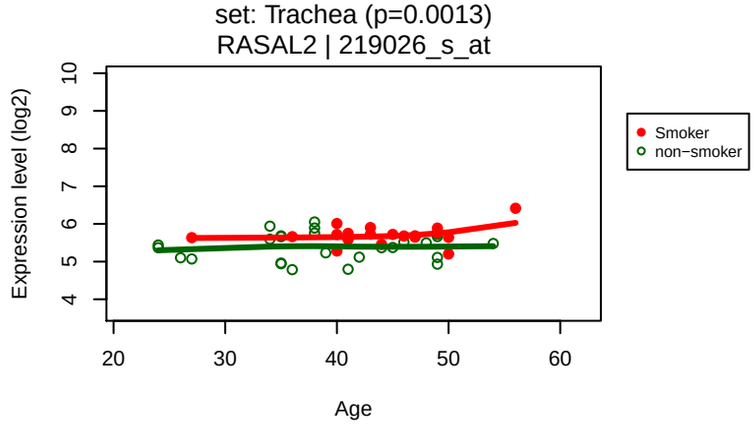

Gene expression ratio in Males
gene: RASAL2 | 219026_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7681
p=0.456
p=0.0139
p=0.4921
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASAL2 | 219026_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7553
p=0.1186
p=0.2233
p=0.9075
SmallAirway2
LargeAirway2
Trachea
Macrophage2

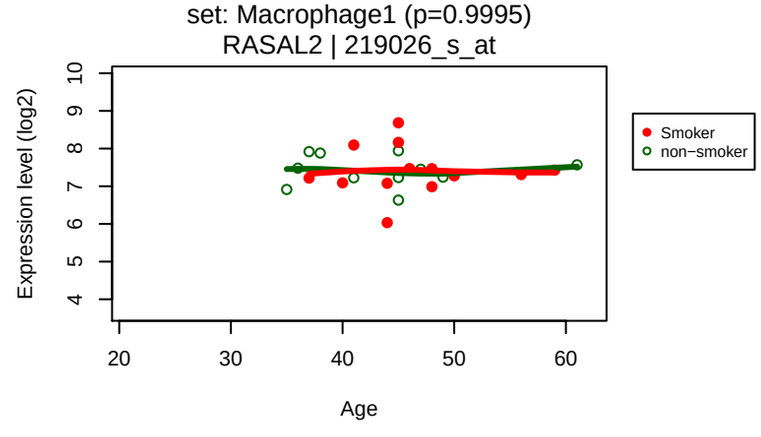

set: SmallAirway1 (p=0.8307)
RASAL2 | 219026_s_at
Expression level (log2)
Age
Smoker
non−smoker

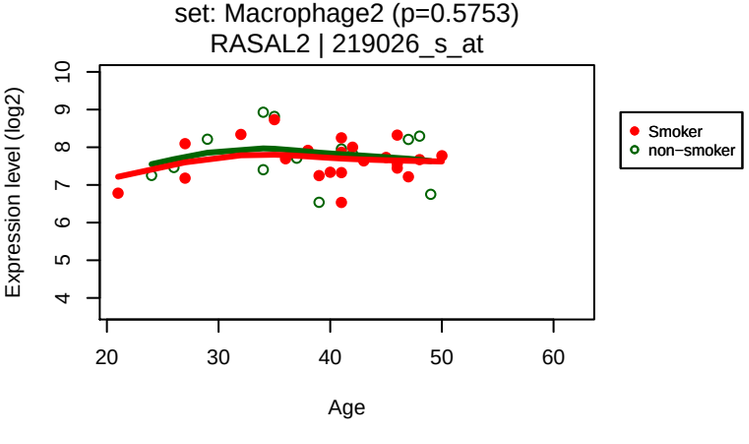

set: SmallAirway2 (p=0.8876)
RASAL2 | 219026_s_at
Expression level (log2)
Age
Smoker
non−smoker

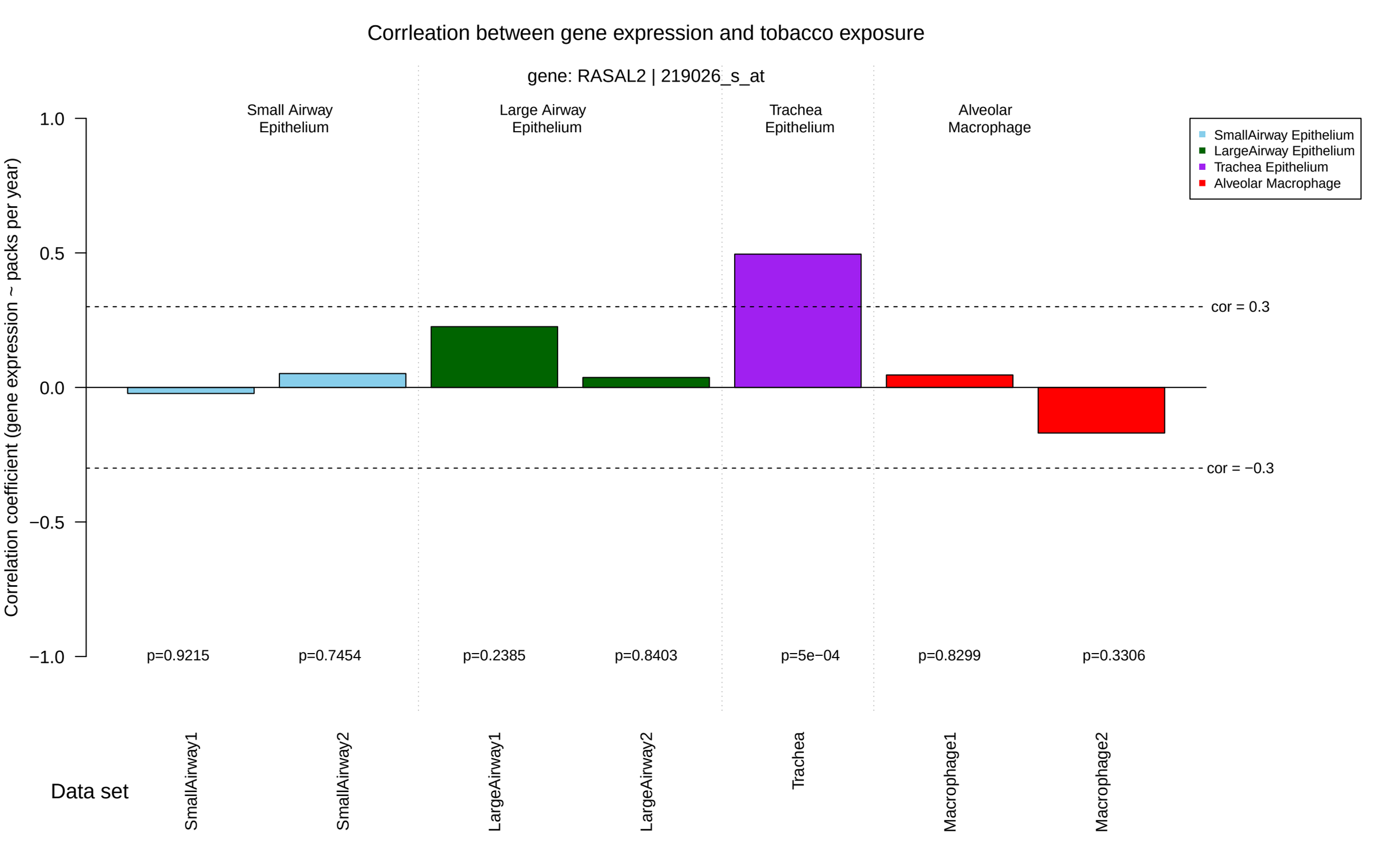

set: LargeAirway1 (p=0.6232)
RASAL2 | 219026_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8612)
RASAL2 | 219026_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0013)
RASAL2 | 219026_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9995)
RASAL2 | 219026_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5753)
RASAL2 | 219026_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASAL2 | 219026_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9215
p=0.7454
p=0.2385
p=0.8403
p=5e−04
p=0.8299
p=0.3306
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2